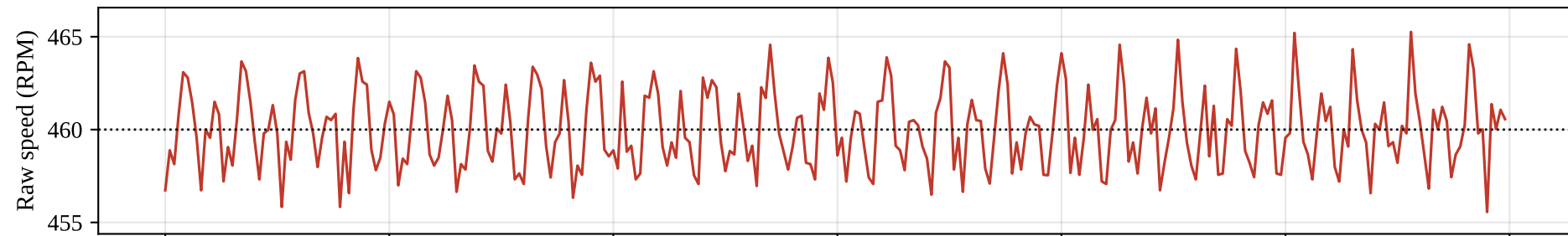
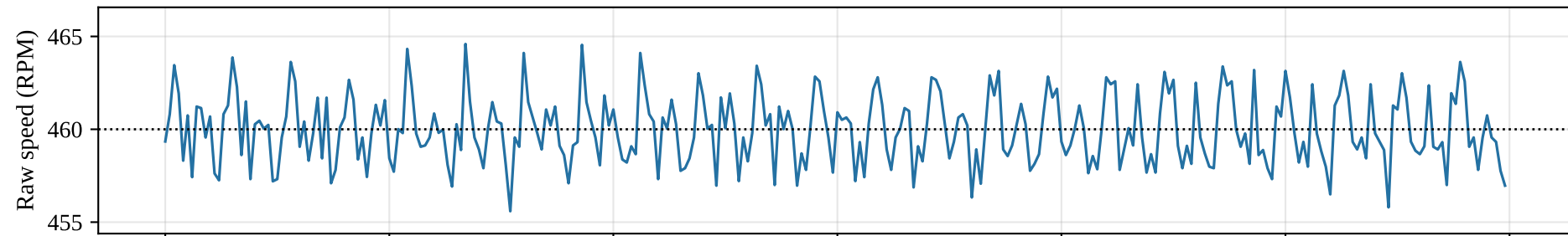


Raw motor speed at steady state, 460 RPM target (same vertical scale on all panels)

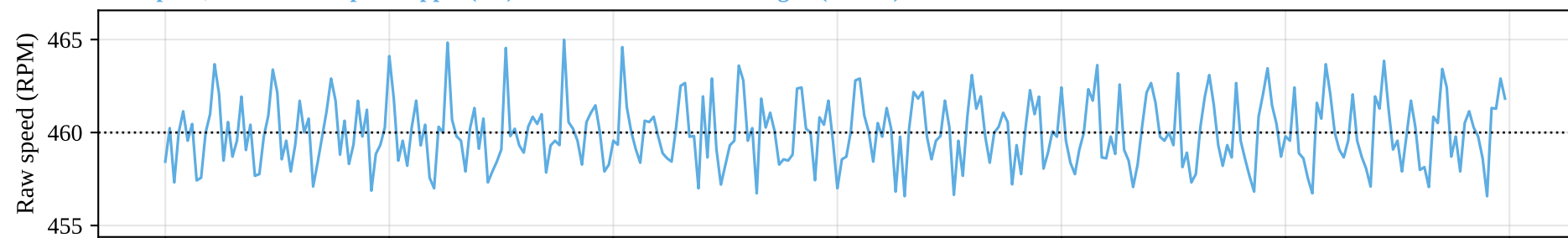
A: baseline (Kp 1.56, Ki 41.2) → speed ripple (std) 2.17 RPM = 0.47 % of target



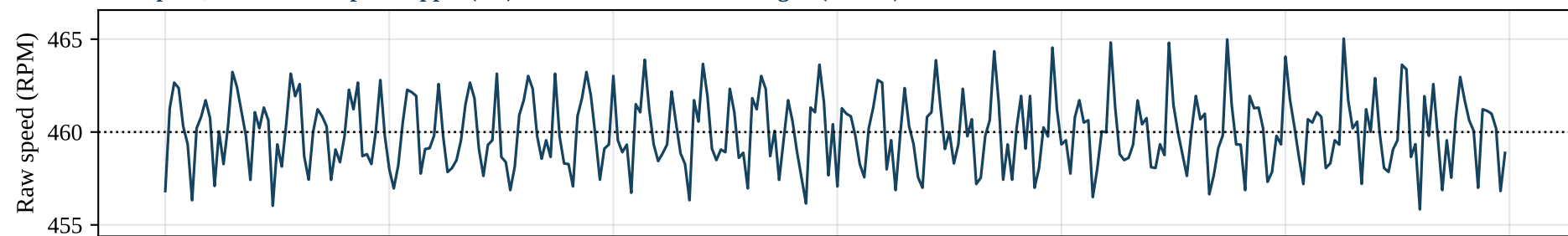
C3: Kp 0.8 → speed ripple (std) 1.93 RPM = 0.42 % of target (-11 %)



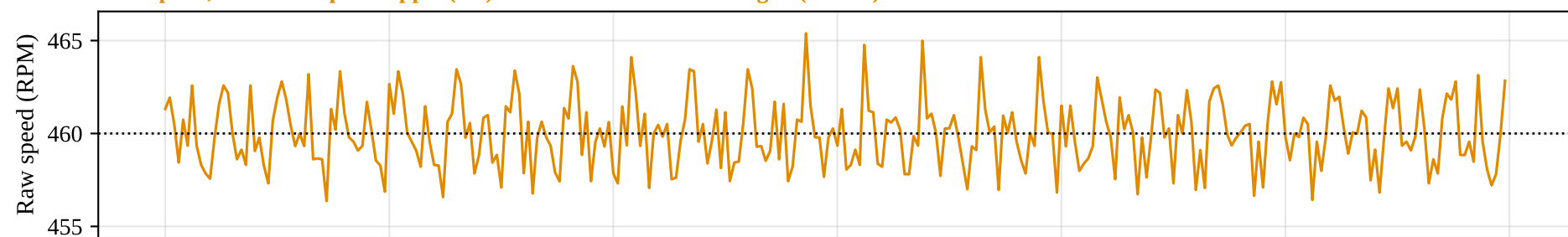
F2: Kp 0.8, Ki 25.0 → speed ripple (std) 1.87 RPM = 0.41 % of target (-14 %)



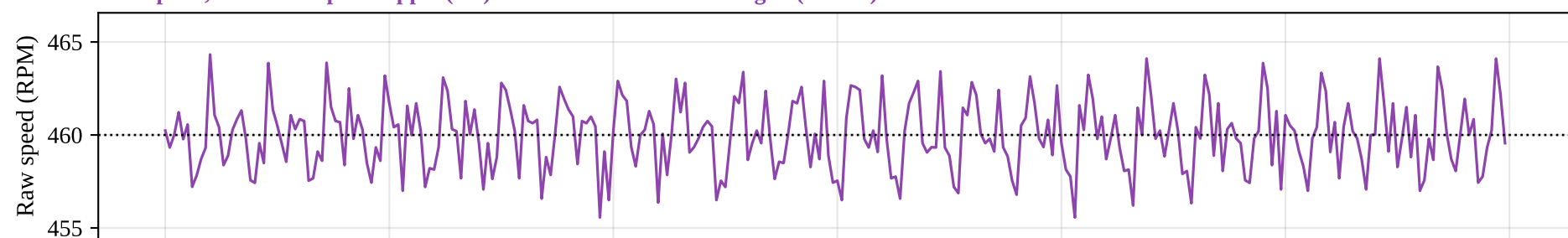
F4: Kp 0.8, Ki 15.0 → speed ripple (std) 1.90 RPM = 0.41 % of target (-12 %)



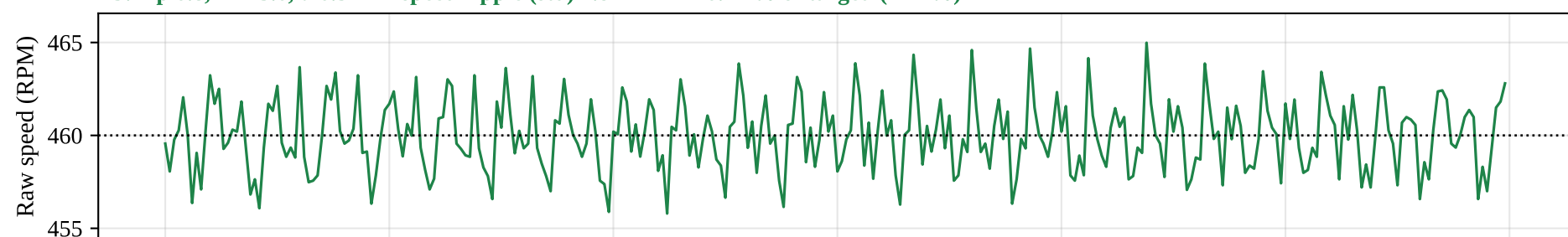
D3: Kp 1.0, α 0.5 → speed ripple (std) 1.85 RPM = 0.40 % of target (-15 %)



F1: Kp 0.8, α 0.5 → speed ripple (std) 1.88 RPM = 0.41 % of target (-14 %)



F3: Kp 0.8, Ki 25.0, α 0.5 → speed ripple (std) 1.87 RPM = 0.41 % of target (-14 %)



10.0 10.5 11.0 11.5 12.0 12.5 13.0
Time (s)